Seng 401 Test Suites

Scripted Testing

Signup Test Cases

TC001	**Successful Signup**	
Preconditions	**User must be on the Signup page**	
Procedure	**Action**	**Expected Result**
1	Enter a valid username	Username is shown in respective field
2	Enter correct password	Password is shown in respective field
3	Click the Signup button	System authenticates user and takes them to the home page
Expected Result	1. User is re-directed to the main menu 2. Main menu is displayed	
Postcondition	User is logged in	

TC002	**Signup with Existing Username**	
Preconditions	**User must be on the Signup page. An existing user account already exists.**	
Procedure	**Action**	**Expected Result**
1	Enter an existing username	Username is shown in respective field

2	Enter a valid password	Password is shown in respective field
3	Click the Signup button	System displays error message “Username already exists”
Expected Result	1. Error message appears 2. User remains on Signup page	
Postcondition	User not logged in and remains on Signup page	

TC003	**Signup with Incorrect Password Confirmation**	
Preconditions	**User must be on the Signup page**	
Procedure	**Action**	**Expected Result**
1	Enter a valid username	Username is shown in respective field
2	Enter a password	Password is shown in respective field
3	Enter a different password in confirm field	Confirm password in respective field
4	Click the Signup button	System displays error message: “Passwords do not match”
Expected Result	1. Error message displayed 2. User remains on Signup page	
Postcondition	User not signed up and remains on Signup page	

TC004	Signup withNo Username	
Preconditions	**User must be on the signup page**	
Procedure	**Action**	**Expected Result**
1	Leave username field empty	No username entered
2	Enter a valid password	Password is shown in respective field
3	Click the Signup button	System displays error message: “Please fill in all fields”
Expected Result	1. Error message appears 2. User remains on Signup page	
Postcondition	User not signed up and remains on Signup page	

TC005	Signup with No Password	
Preconditions	**User must be on the Signup page**	
Procedure	**Action**	**Expected Result**
1	Enter a valid username	Username is shown in respective field
2	Leave password field empty	No password entered
3	Click the Signup button	System displays error message: “Password required”
Expected Result	1. Error message appears 2. User remains on Signup page	

Postcondition	User not signed up and remains on Signup page

TC006	**Signup with Special Characters in Username**	
Preconditions	**User must be on the Signup page**	
Procedure	**Action**	**Expected Result**
1	Enter username with special characters	Username is shown in respective field
2	Enter a valid password	Password is shown in respective field
3	Click the Signup button	System accepts or rejects based on validation rules; appropriate message displayed
Expected Result	If accepted: User is redirected to main menu. If rejected: Error message appears explaining valid characters.	
Postcondition	User is either logged in or remains on Signup page based on validation	

Login Test Cases

TC007	**Successful Login**	
Preconditions	**User must be on the Login page. A valid user account exists**	
Procedure	**Action**	**Expected Result**
1	Enter a valid username	Username is shown in respective field
2	Enter correct	Password is shown in respective field

	password	
3	Click the Login button	System authenticates user and takes them to the home page
Expected Result	1. User is re-directed to the main menu 2. Main menu is displayed	
Postcondition	User is logged in	

TC008	**Login with Incorrect Password**	
Preconditions	**User must be on the login page**	
Procedure	**Action**	**Expected Result**
1	Enter a valid username	Username is shown in respective field
2	Enter incorrect password	Password is shown in respective field
3	Click the Login button	System displays error message: “Invalid username or password”
Expected Result	1. Error message appears 2. User remains on login page	
Postcondition	User not logged in and remains on login page	

TC009	**Login with No Username**	
Preconditions	**User must be on the login page**	
Procedure	**Action**	**Expected Result**
1	Leave username	No username entered

	field empty	
2	Enter a password	Password is shown in respective field
3	Click the Login button	System displays error message: “Username required”
Expected Result	1. Error message displayed 2. User remains on Login page	
Postcondition	User not logged in and remains on login page	

TC0010	**Login with No Password**	
Preconditions	**User must be on the login page**	
Procedure	**Action**	**Expected Result**
1	Enter a valid username	Username is shown in respective field
2	Leave password field empty	No password entered
3	Click the Login button	System displays error message: “Password required”
Expected Result	1. Error message displayed 2. User remains on Login page	
Postcondition	User not logged in and remains on login page	

TC0011	**Login with No Username and No Password**
Preconditions	**User must be on the login page**

Procedure	Action	Expected Result
1	Leave username field empty	No username entered
2	Leave password field empty	No password entered
3	Click the Login button	System displays error message: “Username and password required”
Expected Result	1. Error message displayed 2. User remains on Login page	
Postcondition	User not logged in and remains on login page	

TC0012	**Login with Invalid Username**	
Preconditions	**User must be on the login page**	
Procedure	**Action**	**Expected Result**
1	Enter an invalid/non-existent username	Username is shown in respective field
2	Enter any password	Password is shown in respective field
3	Click the Login button	System displays error message: “Invalid username or password”
Expected Result	1. Error message displayed 2. User remains on Login page	
Postcondition	User not logged in and remains on login page	

Main Menu / New Game Test Cases

TC0013	**Start New Game**	
Preconditions	**User must be on the Main menu and logged in**	
Procedure	**Action**	**Expected Result**
1	Select “New Game” option	System initializes a new game
Expected Result	1. User is re-directed to the game screen 2. New game begins with initial map, currency, and timer 3. Tutorial messages are displayed (if applicable)	
Postcondition	User is in an active game session	

TC0014	**Access Settings from Main Menu**	
Preconditions	**User must be on the Main menu and logged in**	
Procedure	**Action**	**Expected Result**
1	Select “Settings” option	System displays settings menu
Expected Result	1. Settings options are displayed (e.g., audio, controls) 2. User can modify and save settings	
Postcondition	User remains on settings or returns to main menu	

TC0015	Logout from Main Menu	
Preconditions	**User must be on the Main menu and logged in**	
Procedure	**Action**	**Expected Result**
1	Select “Logout” option	System logs user out
Expected Result	1. User is re-directed to Login page 2. Session is terminated	
Postcondition	User is logged out and must log in again to access game	

In-Game Mechanics Test Cases

TC0016	Verify currency increases over time	
Preconditions	**User must be in an active game session**	
Procedure	**Action**	**Expected Result**
1	Note current in-game currency amount	Amount is displayed
2	Wait for 10 seconds without taking any action	Currency amount increases periodically

3	Compare new amount with original	New amount is greater than original
Expected Result	In game currency increases over time automatically	
Postcondition	Currency continues to increase over time	

TC0017	**Verify currency display updates correctly**	
Preconditions	**User must be in an active game session**	
Procedure	**Action**	**Expected result**
1	Observe currency display	Currency is displayed in correct format
2	Perform action that costs currency	Currency display updates immediately to reflect new amount
Expected Result	Currency is displayed properly and updates in real-time after actions	
Postcondition	Currency display remains accurate throughout gameplay	

TC0018	**Select Firefighter from City**	
Preconditions	**User must be in an active game session. City must have at least 1 firefighter available.**	
Procedure	**Action**	**Expected Result**

1	Click on a city to view its info	City info displays number of available firefighters
2	Select a firefighter from that city	Firefighter sprite appears and is selectable. City's firefighter count decreases by 1
Expected Result	1. Firefighter sprite moves in the correct direction when selected 2. Firefighter count from city updates accordingly	
Postcondition	Firefighter is selected and can be deployed	

TC0019	**Deploy firefighter to put out fire**	
Preconditions	**User must be in an active game session. A fire must be present on the map. At least one firefighter must be selected.**	
Procedure	**Action**	**Expected Result**
1	Select a firefighter	Firefighter is ready for deployment
2	Click on a tile that is on fire	Firefighter moves to the fire tile
3	Wait for firefighting action to complete	Fire is extinguished
Expected Result	Fire is put out after firefighter reaches the tile	
Postcondition	Fire tile is now in burnt state	

TC0020	Verify burnt tile changes after fire extinguished	
Preconditions	**User must be in an active game session. A fire must be present**	
Procedure	**Action**	**Expected Result**
1	Identify a tile that is on fire	Tile shows fire animation/indicator
2	Deploy a firefighter to that tile	Firefighter move to tile
3	Observe tile after fire is put out	Tile style changes from normal/burning to burnt
Expected Result	Tile style changes to burnt after fire is extinguished	
Postcondition	Tile remains burnt until recovery	

TC0021	Verify Burnt Tile recovers after 30 seconds	
Preconditions	**User must be in an active game session. A burnt tile must exist. Level must increase (or recovery mechanic triggers)**	
Procedure	**Action**	**Expected Result**
1	Note a burnt tile	Tile appears burnt
2	Wait for 30 seconds after level increases	Tile recovers from burnt to normal state

Expected Result	After 30 seconds (post-level increase), burnt tiles return to normal
Postcondition	Tile is normal and usable again

TC0022	**Verify City Burned Ends Game**	
Preconditions	**User must be in an active game session. A city must be at risk of burning**	
Procedure	**Action**	**Expected Result**
1	Allow fires to spread to a city	City burns
2	Observe game state	Game ends immediately
Expected Result	When a city is burned, the game ends	
Postcondition	Game over screen is displayed	

TC0023	**Verify fire spawn location rules**	
Preconditions	**User must be in an active game session. Fire spawn event occurs.**	
Procedure	**Action**	**Expected Result**
1	Observe location of new fire	Fire does not spawn directly on a building or city or location where fire cannot spawn (e.g. lake, mountains etc.)

2	Check proximity to buildings	Fire is not immediately adjacent to buildings (or within defined safe distance)
Expected Result	Fire appears at valid location away from buildings	
Postcondition	Fire is present and can be addressed	

TC0024	**Navigate across map**	
Preconditions	**User must be in an active game session.**	
Procedure	**Action**	**Expected Result**
1	Click and drag the map or use navigation controls	Map view shifts
2	Scroll or zoom (if applicable)	Map zooms in/out
3	Move to different sections of the map	All parts of the map become viewable
Expected Result	User can navigate across the map and view different areas	
Postcondition	Map view remains at last position unless reset	

TC0025	**Verify City info Display**
Preconditions	**User must be in an active game session.**

Procedure	Action	Expected Result
1	Click on a city	City info panel appears
2	Verify displayed information	City name, number of firefighters, ability to enact policy are displayed
Expected Result	Each city displays its info correctly (firefighter count, policy options)	
Postcondition	City info is accessible for all cities	

TC0026	**Enact Policy in City**	
Preconditions	**User must be in an active game session. City must have policy available. User must have sufficient currency.**	
Procedure	**Action**	**Expected Result**
1	Click on a city	City info is displayed
2	Select “Enact Policy” option	Policy is applied to the city
3	Verify currency deduction	Currency decreases by policy cost
4	Verify policy effect	City gains policy benefits (e.g. more firefighters, fire resistance)
Expected Result	Policy is enacted, currency adjusts, and city status updates	
Postcondition	Policy remains active on city until disabled	

TC0027	Pause Game	
Preconditions	**User must be in an active game session.**	
Procedure	**Action**	**Expected Result**
1	Click “Pause” button	Game timer stops
2	Verify game state	Fires do not spread, currency does not increase, actions cannot be taken (or are limited)
Expected Result	Game is paused. User can see pause menu options.	
Postcondition	Game remains paused until resumed	

TC0028	Save and Exit from Pause Menu	
Preconditions	**User must be in an active game session and have paused the game**	
Procedure	**Action**	**Expected Result**
1	Click “Pause”	Pause menu appears
2	Select “Save and Exit”	Game state is saved
3	Verify user is taken to main menu	User is redirected to main menu

Expected Result	1. Game saves successfully 2. User returns to main menu
Postcondition	Saved game file exists and can be loaded

TC0029	**Resume from Pause**	
Preconditions	**User must be in an active game session and have paused the game**	
Procedure	**Action**	**Expected Result**
1	Click “Pause”	Pause menu appears
2	Click “Resume”	Game timer resumes
3	Verify game state	Game continues from exactly where it was paused
Expected Result	Game resumes correctly with all states preserved	
Postcondition	Game session is active	

Load Game Test Cases

TC0030	**Load existing Saved Game**
Preconditions	**User must be on the Main menu. A saved game file must exist.**

Procedure	Action	Expected Result
1	Select “Load Game” option	Game that was previously saved is loaded
Expected Result	1. Game loads successfully 2. User is taken to the game screen	
Postcondition	Loaded game session is active	

TC0031	**Verify currency persists after load**	
Preconditions	**User has a saved game with a specific currency amount. Saved game is loaded.**	
Procedure	**Action**	**Expected Result**
1	Note currency amount before saving	Amount is recorded
2	Save and exit game	Game saved
3	Load the saved game	Game loads
4	Compare loaded currency with saved amount	Currency amount is identical to before save
Expected Result	In game currency is the same as before save	
Postcondition	Currency persists correctly across save/load	

TC0032	Verify firefighter count persists after load	
Preconditions	**User has saved a game with specific firefighter counts per city. Saved game is loaded.**	
Procedure	**Action**	**Expected Result**
1	Note firefighter count in each city before saving	Counts are recorded
2	Save and exit game	Game saved
3	Load the saved game	Game loads
4	Compare loaded firefighter counts	Counts are identical to before save
Expected Result	Firefighter counts are the same as before save	
Postcondition	Firefighter counts persist correctly across save/load	

TC0033	Verify Policies persist After Load	
Preconditions	**User has enacted policies on cities before saving. Saved game is loaded.**	
Procedure	**Action**	**Expected Result**
1	Enact policies on one or more cities	Policies are active

2	Save and exit game	Game saved
3	Load the saved game	Game loads
4	Check city policies	Policies are still enacted and active
Expected Result	Policies enacted before save remain active after load	
Postcondition	Policies persist correctly across save.load	

TC0034	**Verify Game Level Persists After Load**	
Preconditions	**User has saved a game at a specific level. Saved game is loaded.**	
Procedure	**Action**	**Expected Result**
1	Note current game level before saving	Level is recorded
2	Save and exit game	Game save
3	Load the saved game	Game loads
4	Compare loaded level	Level is the same as before save
Expected Result	Game level is the same as before save	

Postcondition	Level persists correctly across save/load

TC0035	**Verify Map State Persists After Load**	
Preconditions	**User has saved a game with specific map conditions (burnt tiles, fire locations). Saved game is loaded.**	
Procedure	**Action**	**Expected Result**
1	Note map state (fire locations, burnt tiles) before saving	State is recorded
2	Save and exit game	Game saved
3	Load the saved game	Game loads
4	Compare map state	Map state is identical to before save
Expected Result	Game map (tile states, fire locations) is the same as before save	
Postcondition	Map state persists correctly across save/load	

TC0036	**Verify Timer and Level progress Persists After Load**	
Preconditions	**User has saved game at a specific timer value (e.g. 15 seconds into the level). Saved game is loaded**	
Procedure	**Action**	**Expected Result**

1	Note timer value and level progress before saving	Values are recorded
2	Save and exit game	Game saved
3	Load the saved game	Game loaded
4	Compare timer and progress	Timer and level progress are the same as before save
Expected Result	Timer and level progress are the same as before save; level has not restarted	
Postcondition	Timer and progress persist correctly across save/load	

TC0037	**Verify fires do not spread while game was saved**	
Preconditions	**User has a saved game with fires present. Saved game is loaded after a period of time**	
Procedure	**Action**	**Expected Result**
1	Save games with fires at specific locations	Gane saved
2	Wait for a period of time (e.g. 2 minutes) without playing	No game activity

3	Load the saved game	Game loads
4	Fires have not spread in the absence of the user	Fire locations are exactly the same as when saved; no spread occurred
Expected Result	Fires have not spread in the absence of the user	
Postcondition	Game resumes exactly as it was saved	

TC0038	**Verify all functionalities work after Load**	
Preconditions	**Saved game is loaded successfully.**	
Procedure	**Action**	**Expected Result**
1	Deploy a firefighter	Firefighter moves and puts out fire
2	Enact a policy	Policy is applied, currency decreases
3	Pause the game	Game pauses
4	Resume and save the game	Game saves successfully
Expected Result	All functionalities (deploy firefighters, enact policies, pause, save) work identically to a fresh new game	
Postcondition	Loaded game behaves exactly like a non-loaded game	

Game Over / End Conditions Test Cases

TC0039	**Game Over screen Display**	
Preconditions	**Game ending condition must be triggered**	
Procedure	**Action**	**Expected Result**
1	Allow a city to burn	Game ends
2	Observe screen	Game over screen is displayed
Expected Result	Game over screen is displayed	
Postcondition	Game is no longer active	

TC0040	**Return to Home Screen after Game Over**	
Preconditions	**Game Over screen is displayed**	
Procedure	**Action**	**Expected Result**
1	Click “Main Menu” or “OK” button on game over screen	User is redirected to home screen (main menu)
Expected Result	User is taken back to the home screen	
Postcondition	User is on main menu and can start a new game or load a saved game	

Tutorial Test Cases

TC0022	**Tutorial Messages Display Correctly**	
Preconditions	**User starts a new game for the first time**	
Procedure	**Action**	**Expected Result**
1	Start new game	Tutorial messages appear at appropriate times
2	Follow tutorial prompts	Messages guide user through basic actions
3	Complete tutorial actions	Messages update to next step
Expected Result	Tutorials are working properly and messages are displayed correctly	
Postcondition	User completes tutorials or can skip	

Music Test Cases

TC0042	**Music plays on game launch**	
Preconditions	**User launches the game. Music volume is not muted**	
Procedure	**Action**	**Expected Result**
1	Launch the game application	Game loads and music begins playing
2	Verify music plays on login screen	Background music is audible

Expected Result	Music starts automatically upon game launch and continues on login screen
Postcondition	Music continues playing throughout menus

TC0043	**Music Transition between Scenes**	
Preconditions	**User is logged in. Music is enabled**	
Procedure	**Action**	**Expected Result**
1	Observe music on main menu	Main menu music plays
2	Start a new game	Music transitions to in-game music smoothly
3	Pause the game	Music continues or transitions to pause music (if applicable)
4	Return to main menu	Music transitions back appropriately
Expected Result	Music transitions smoothly between screens without audio glitches	
Postcondition	Appropriate music plays for each screen/game state	

TC0044	**Music persists during gameplay**	
Preconditions	**User must be in an active game session. Music is enabled.**	
Procedure	**Action**	**Expected Result**

1	Play game for extended period	Music continues playing without interruption
2	Perform various actions (deploy firefighters, enact policies)	Music does not cut out or glitch during actions
Expected Result	Music plays continuously and consistently during gameplay	
Postcondition	Music continues until game ends or user mutes	

TC0045	**Mute Music from settings**	
Preconditions	**User is on main menu or in-game. Music is playing**	
Procedure	**Action**	**Expected Result**
1	Open settings menu	Settings options are displayed
2	Toggle Music to OFF/Mute	Music stops immediately
3	Exit settings and continue gameplay	No music plays
Expected Result	Music mutes immediately when setting is toggled off	
Postcondition	Music remains muted until re-enabled	

TC0046	Unmute Music from settings	
Preconditions	**User has previously muted music. Settings menu is open.**	
Procedure	**Action**	**Expected Result**
1	Toggle music to ON/Unmute	Music resumes playing
2	Verify music plays at previous volume level	Volume is consistent with last saved setting
Expected Result	Music resumes correctly when unmuted	
Postcondition	Music continues playing until muted again	

TC0047	Music volume adjustment	
Preconditions	**User is in settings menu. Music is playing**	
Procedure	**Action**	**Expected Result**
1	Locate music volume slider	Slider is displayed
2	Adjust music volume up and down	Music volume changes in real-time
3	Set volume to 0%	Music is silent (equivalent to mute)
4	Set volume to 100%	Music is at maximum volume

Expected Result	Volume slider controls music volume accurately across full range
Postcondition	Volume setting is saved for future sessions

TC0048	**Music Volume persists after Restart**	
Preconditions	**User has adjusted music volume to a specific level (e.g. 50%)**	
Procedure	**Action**	**Expected Result**
1	Set music volume to a specific level	Volume is applied
2	Exit the game completely	Game closes
3	Relaunch the game	Music plays at previously set volume level
Expected Result	Music volume setting persists across game sessions	
Postcondition	User preferences are saved	

TC0049	**Music does not overlap with sound effects**	
Preconditions	**User must be in an active game session. Music and sound effects are enabled**	
Procedure	**Action**	**Expected Result**

1	Play music at moderate volume	Music is audible
2	Perform actions with sound effects (e.g. deploy firefighter, click buttons)	Sound effects play at appropriate volume without drowning out music
3	Verify both are audible simultaneously	Music and sound effects mix properly
Expected Result	Music and sound effects play concurrently without one overpowering the other	
Postcondition	Audio mixing remains balanced	

TC0050	**Music fades out on game over**	
Preconditions	**User must be in an active game session. Music is playing**	
Procedure	**Action**	**Expected Result**
1	Allow game to end	Music fades out (or transitions to game over music)
2	Observer game over screen	Game over music plays or music stops appropriately
Expected Result	Music handles game over state correctly (fade out or change)	
Postcondition	Game over state has appropriate audio	

Fire Mechanics Test Cases

TC0051	**Fire spreads over time**	
Preconditions	**User must be in an active game session. A fire is present on the map**	
Procedure	**Action**	**Expected Result**
1	Identify location of a fire	Fire location is noted
2	Wait for 10-15 seconds without extinguishing	Fire spreads to adjacent tiles
3	Verify spread pattern	Fire spreads logically to neighboring tiles
Expected Result	Fires spread over time if not extinguished	
Postcondition	Fires may grow larger if left unattended	

TC0052	**Multiple fires can exist**	
Preconditions	**User must be in an active game session.**	
Procedure	**Action**	**Expected Result**
1	Allow time to pass for fire spawn events	Multiple fires appear on map

2	Verify each fire location	Each fire is independent and spreads individually
Expected Result	Game can handle multiple fires at once	
Postcondition	Multiple fires can be managed simultaneously	

Firefighter Limits Test Cases

TC0053	**Cannot Deploy FireFighter if none available**	
Preconditions	**User must be in an active game session. A city has 0 firefighters available.**	
Procedure	**Action**	**Expected Result**
1	Click on a city with - firefighters	City info shows 0 firefighters
2	Attempt to select a firefighter	System prevents selection (grayed out or error message)
Expected Result	User cannot select a firefighter from a city with none available	
Postcondition	User must wait for firefighters to replenish or use another city	

TC0054	**Firefighters Replenish Over time**
Preconditions	**User must be in an active game session. A city has less than maximum firefighters**

Procedure	Action	Expected Result
1	Note current firefighter count in a city	Count is recorded
2	Wait for a period of time	Firefighter count increases
3	Verify replenishment	Count is higher than before
Expected Result	Firefighters replenish over time (or based on game mechanics)	
Postcondition	City gradually regains firefighting capacity	

Policy Effects Test Cases

TC0055	**Policy Effect Applies Correctly**	
Preconditions	**User must be in an active game session. User has sufficient currency.**	
Procedure	**Action**	**Expected Result**
1	Note city's current attributes before policy	Attributes (e.g. firefighter count) are recorded
2	Enact a policy on the city	Currency is deducted

3	Verify policy effect	City attributes change according to policy description
Expected Result	Policy effects are applied correctly to the city	
Postcondition	City benefits from policy until policy expires or is removed	

TC0056	**Cannot Enact Policy without Sufficient Currency**	
Preconditions	**User must be in an active game session. Policy cost exceeds current currency**	
Procedure	**Action**	**Expected Result**
1	Attempt to enact a policy on a city	Enact button is disabled or error message appears
2	Verify currency does not go negative	Currency remains unchanged
Expected Result	User cannot enact policy when currency is insufficient	
Postcondition	User must earn more currency to enact policy	

Level Progression Test Cases

TC0022	**Game level Increases Over time**

Preconditions	User must be in an active game session.	
Procedure	**Action**	**Expected Result**
1	Note current game level	Level is displayed
2	Play game until timer/condition triggers level increase	Level increases
3	Verify increased difficulty	Fire frequency, fire spread rate, other mechanics become harder
Expected Result	Game level increases after set time or conditions are met	
Postcondition	Higher level introduces greater challenge	

TC0058	Burnt Tiles Recover After Level Increase	
Preconditions	**User must be in an active game session. Burnt tiles exist. Level increases**	
Procedure	**Action**	**Expected Result**
1	Ensure burnt tiles are present	Tiles appear burnt
2	Wait for level to increase	Level increases

3	Observe burnt tiles after 30 seconds	Burnt tiles recover to normal state
Expected Result	After level increases, burnt tiles recover after 30 seconds	
Postcondition	Recovered tiles are usable again	

Map Interaction Test Cases

TC0059	**Click and Drag Map Navigation**	
Preconditions	**User must be in an active game session. Map is larger than viewport**	
Procedure	**Action**	**Expected Result**
1	Click and hold on map or use keyboard buttons	Map is grabbed or moved
2	Drag in any direction if using click	Map scrolls smoothly in direction of drag
3	Release click or buttons	Map stops scrolling/moving
Expected Result	User can navigate across the map by click or keyboard buttons	
Postcondition	Map view is updated	

TC0060	Zoom In/Out functionality	
Preconditions	**User must be in an active game session. Zoom feature is implemented**	
Procedure	**Action**	**Expected Result**
1	Zoom is using scroll method	Map view zooms in, showing more detail on fewer tiles
2	Zoom out	Map view zooms out, showing more tiles
3	Verify interactions at different zoom levels	Clicking on cities.firefighters works at all zoom levels
Expected Result	User can zoom in and out to view map at different scales	
Postcondition	Zoom level persists until changed	

UI and Display Test Cases

TC0061	Timer Display Updates Correctly	
Preconditions	**User must be in an active game session.**	
Procedure	**Action**	**Expected Result**
1	Observe timer display	Timer counts up or down correctly
2	Pause the game	Timer stops

3	Resume the game	Timer resumes from paused value
Expected Result	Timer is displayed properly and updates accurately	
Postcondition	Time reflects actual game time	

TC0062	**City info Updates After Firefighter Deployment**	
Preconditions	**User must be in an active game session. City has firefighters**	
Procedure	**Action**	**Expected Result**
1	Open city info panel and note firefighter count	Count is displayed
2	Select firefighter from that city	Firefighter count in info panel decreases by 1 immediately
3	Close and reopen the panel	Count remains decreased
Expected Result	City info updates in real-time after actions	
Postcondition	All city displays are accurate	

Edge Cases

TC0063	**Game Handles Rapid Clicking/Actions**	
Preconditions	**User must be in an active game session.**	
Procedure	**Action**	**Expected Result**
1	Rapidly click on multiple cities	Game registers actions without crashing
2	Rapidly deploy multiple firefighters	Deployments queue or process correctly without crashing the game
3	Rapidly pause/unpause	Game handles state changes without freezing
Expected Result	Game remains stable under rapid input conditions	
Postcondition	Game remains consistent	

TC0064	**Game handles Low System Resources**	
Preconditions	**Game is running system with limited resources (or simulate with background processes)**	
Procedure	**Action**	**Expected Result**
1	Run game with limited memory/CPU	Game does not crash

2	Play normally	Performance may degrade but game remains functional
3	Save and load game	Save/load functions still work
Expected Result	Game handles low resource conditions gracefully (graceful degradation)	
Postcondition	Game may warn user of performance issue	